**Course Code : ECT 451-3** **UVXV/MW – 24 / 1664**

# Seventh Semester B. Tech. ( Electronics and Communication Engineering ) Examination

## DIGITAL IMAGE AND VIDEO PROCESSING

Time : 3 Hours ] [ Max. Marks : 60

**Instructions to Candidates :—**

(1) All questions carry marks as indicated.
(2) Assume suitable data wherever necessary.
(3) Illustrate your answers with neat sketches.
(4) Use of Non – programmable calculator is permitted.

1. (a) Assume a person 1.4 meters in height as shown in Fig. 1a is viewed by an observer from a distance of 5 meters, calculate the height of retinal image.

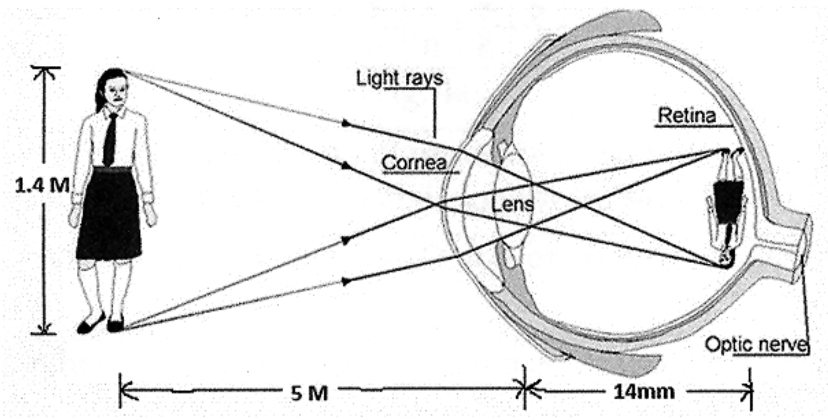


Fig. 1a

3(CO1)

(b) Consider an image segment shown below in Fig. Q.1b and compute the distance $D_8$ and $D_e$ between pixel p and q. Show the shortest – 4, shortest–8 and shortest–m paths between pixels p and q where, V = {80, 83, 102}. If a certain path doesn't exist explain its reason.

[Table On Next page]

| | | | | |
|---|---|---|---|---|
| 83 | 102 | 80 | 102 | 80 |
| 80 | 83 | 80 | 84 | 102(q) |
| 102 | 102 | 83 | 91 | 84 |
| 83(p) | 80 | 84 | 102 | 91 |
| 91 | 102 | 80 | 83 | 84 |

Fig. Q.1b 7(CO1)

2. (a) Verify that 4x4 DFT coefficient matrix satisfy unitary function. 3(CO3)

(b) Apply Haar wavelet transform to a sample image segment given below in Fig. 2b to achieve level–1 decomposition.

| | | | |
|---|---|---|---|
| 1 | 5 | 6 | 0 |
| 2 | 6 | 4 | 3 |
| 5 | 9 | 7 | 2 |
| 7 | 6 | 9 | 8 |

Fig. 2b 7(CO3)

3. (a) Apply bit plane slicing technique to the image in Fig. 3a having intensity values from 0 to 15. Construct the image by eliminating plane(s) retaining most of the image information. Comment on the compression achieved.

| | | | |
|---|---|---|---|
| 12 | 4 | 4 | 4 |
| 8 | 12 | 0 | 0 |
| 6 | 12 | 6 | 6 |
| 12 | 8 | 8 | 6 |

Fig.3a 6(CO3)

(b) Discuss the need of image compression and explain spatial redundancy, coding redundancy and compression ratio. 4(CO3)

4. (a) Apply the histogram equalization on the 4–bit image shown in Fig.4a and compare the original and equalized histogram.

| | | | | |
|---|---|---|---|---|
| 0 | 8 | 7 | 0 | 5 |
| 6 | 7 | 9 | 8 | 6 |
| 5 | 5 | 9 | 5 | 7 |
| 5 | 5 | 6 | 8 | 5 |
| 0 | 0 | 0 | 8 | 6 |

Fig.4a 6(CO3)

(b) Perform a comparison of chrominance subsampling formats mentioning their specifications and suggest subsampling for DVD. 4(CO2)

5. (a) Describe sampling of video in 2D for progressive and interlaced scans. 6(CO2)

(b) Discuss general methodologies in motion estimation and the aperture problem. 4(CO4)

6. (a) Describe Hierarchical Feature Matching Motion Estimation Scheme (HFM–ME). 5(CO4)

(b) Develop an image processing system which will accept an input gray scale image shown in Fig. Q6b–1 and result in output image shown in Fig.Q.6b–2. Draw the block diagram of the system and explain its functioning you have proposed.

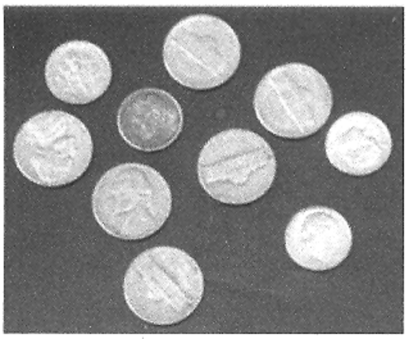

Fig.Q.6b-1

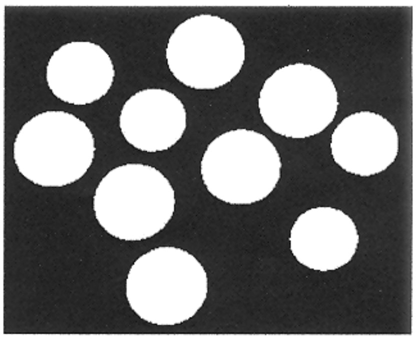

Fig.Q.6b-2

5(CO5)